

ZACHARY SMITH

DATA SCIENCE & MACHINE LEARNING ENGINEER

Manhattan Beach, CA • (310) 890-6248 • zachary90266@gmail.com • github.com/Delicatessen0 • delicatessen0.github.io/Portfolio-General

EDUCATION

Pennsylvania State University | B.S. in Data Science

University Park, PA | 2022 – May 2026

Mira Costa High School | High School Diploma

Manhattan Beach, CA | 2018 – 2022

TECHNICAL SKILLS

- **Languages:** Python, R, SQL, JavaScript (ES6+), HTML5/CSS3
- **AI & Machine Learning:** scikit-learn, XGBoost, PyTorch, TensorFlow, HuggingFace Transformers, FAISS, VADER NLP
- **Web & Frameworks:** Next.js 16, React 19, Tailwind CSS, Recharts, Chart.js, Node.js
- **Data & Developer Tools:** MySQL, SSMS, Tableau, Git/GitHub, VSCode, Pandas, NumPy, Matplotlib

PROFESSIONAL EXPERIENCE

Data Science Intern | Pacific Healthworks

Feb 2023 – Jul 2026

- Structured, analyzed, and visualized healthcare datasets using MySQL and SSMS to optimize hospital operational trends.
- Built interactive Tableau performance dashboards enabling clinical leadership to evaluate resource allocation efficiency.
- Automated data transformation pipelines and reporting tables, reducing manual visualization workflows by over 40%.

Market Analytics Intern | Career Movement

May 2023 – Aug 2023

- Conducted market research and trend analysis on employment data to inform strategic hiring recommendations.
- Performed structured database management and cross-functional data verification to maintain data integrity.

Operations Specialist | Pennsylvania State University

2024 – 2025

- Collaborated within high-volume event teams to execute customer service workflows and inventory tracking.

FEATURED DATA SCIENCE & SOFTWARE PROJECTS

Game Recommendation LLM Engine | Python • HuggingFace • Transformers • FAISS

2026

- Engineered a semantic recommendation system using HuggingFace transformer embeddings to compute catalog similarity.
- Constructed a high-throughput FAISS vector index enabling sub-millisecond similarity search across large datasets.

Steam Game Market Success Predictor | Python • XGBoost • SteamSpy API • VADER NLP

2026

- Developed an end-to-end ML classification model achieving 85.7% accuracy predicting product market success.
- Extracted live API telemetry and executed VADER sentiment analysis on customer review text to engineer model features.

Live-Service Churn Analytics Model | Python • XGBoost • Cox Hazards • Scikit-Learn

2026

- Built a hybrid classification & Cox Proportional Hazards survival model predicting player churn risk from telemetry.

DataPulse BI Analytics Portal | Next.js 16 • React • TypeScript • Recharts • Surge

2026

- Architected a full-stack executive BI dashboard with interactive time-series modeling, retention heatmaps, and CSV export.

REFERENCES

Tom King (310) 740-1008

Nancy Garcia (310) 569-0540